

Problem 1: Thermal Dissipation in a High-Performance Processor The temperature distribution T on a CPU surface is modeled by $T(x, y) = \arctan\left(\frac{y}{x}\right) \cdot \ln(x^2 + y^2)$ for $x > 0$.

- Calculate the partial derivatives T_x and T_y .
- **Scenario Analysis:** If a temperature sensor is placed at $(1, 1)$, in which coordinate direction (x or y) is the temperature increasing more rapidly?

Function: $T(x, y) = \arctan(y/x) \cdot \ln(x^2 + y^2)$

Step-by-Step Solution:

1. **Identify Rule:** We must use the Product Rule: $(uv)' = u'v + uv'$. Let $u = \arctan(y/x)$ and $v = \ln(x^2 + y^2)$.

2. **Differentiate u and v with respect to x :**

- $\frac{\partial u}{\partial x} = \frac{1}{1+(y/x)^2} \cdot \frac{\partial}{\partial x}\left(\frac{y}{x}\right) = \frac{x^2}{x^2+y^2} \cdot \left(-\frac{y}{x^2}\right) = \frac{-y}{x^2+y^2}$
- $\frac{\partial v}{\partial x} = \frac{1}{x^2+y^2} \cdot (2x) = \frac{2x}{x^2+y^2}$

3. **Combine for T_x :** $T_x = \left(\frac{-y}{x^2+y^2}\right) \ln(x^2 + y^2) + \arctan(y/x) \left(\frac{2x}{x^2+y^2}\right)$

4. **Repeat for y :**

- $\frac{\partial u}{\partial y} = \frac{1}{1+(y/x)^2} \cdot \left(\frac{1}{x}\right) = \frac{x^2}{x^2+y^2} \cdot \frac{1}{x} = \frac{x}{x^2+y^2}$
- $\frac{\partial v}{\partial y} = \frac{2y}{x^2+y^2}$

5. **Combine for T_y :** $T_y = \left(\frac{x}{x^2+y^2}\right) \ln(x^2 + y^2) + \arctan(y/x) \left(\frac{2y}{x^2+y^2}\right)$

Analysis: At $(1, 1)$, $\ln(2) \approx 0.693$ and $\arctan(1) = \pi/4$. $T_x = \frac{-0.693+2(\pi/4)}{2} \approx 0.439$; $T_y = \frac{0.693+2(\pi/4)}{2} \approx 1.132$. Since $T_y > T_x$, the temperature rises faster in the y -direction.

Problem 2: Wave Equation Verification in Signal Processing A signal propagating through a medium is represented by the function $u(x, t) = \cos(kx - \omega t) + e^{x-ct}$.

- Find the second-order partial derivatives $\frac{\partial^2 u}{\partial x^2}$ and $\frac{\partial^2 u}{\partial t^2}$.
- **Scenario Analysis:** Determine the specific relationship between constants k, ω, c required for the term e^{x-ct} to satisfy the wave equation $u_{tt} = c^2 u_{xx}$.

Function: $u(x, t) = \cos(kx - \omega t) + e^{x-ct}$

Step-by-Step Solution:

1. **Time Derivatives:**

- $u_t = -\sin(kx - \omega t) \cdot (-\omega) + e^{x-ct} \cdot (-c) = \omega \sin(kx - \omega t) - ce^{x-ct}$
- $u_{tt} = \omega \cos(kx - \omega t) \cdot (-\omega) - ce^{x-ct} \cdot (-c) = -\omega^2 \cos(kx - \omega t) + c^2 e^{x-ct}$

2. **Space Derivatives:**

- $u_x = -\sin(kx - \omega t) \cdot (k) + e^{x-ct} = -k \sin(kx - \omega t) + e^{x-ct}$
- $u_{xx} = -k \cos(kx - \omega t) \cdot (k) + e^{x-ct} = -k^2 \cos(kx - \omega t) + e^{x-ct}$

3. **Substitution into $u_{tt} = c^2 u_{xx}$:** $-\omega^2 \cos(kx - \omega t) + c^2 e^{x-ct} = c^2 [-k^2 \cos(kx - \omega t) + e^{x-ct}]$
 $-\omega^2 \cos(kx - \omega t) + c^2 e^{x-ct} = -c^2 k^2 \cos(kx - \omega t) + c^2 e^{x-ct}$

4. **Conclusion:** The equation holds if and only if $-\omega^2 = -c^2 k^2$, which simplifies to $\omega = ck$.

Problem 3: Error Surface for a Support Vector Machine (SVM) The margin of a classifier is given by the function $f(w, b) = \frac{1}{\sqrt{w_1^2 + w_2^2}} \exp(-(w_1 b)^2)$.

- Find the gradient $\nabla f(w_1, w_2, b)$ with respect to w_1 and b .
- **Scenario Analysis:** If w_1 represents a weight and b represents a bias, explain how an increase in the bias b affects the sensitivity of the margin at the point $(1, 1, 1)$.

Function: $f(w, b) = (w_1^2 + w_2^2)^{-1/2} e^{-(w_1 b)^2}$

Step-by-Step Solution:

1. **Partial w.r.t w_1 (Product Rule):** $\frac{\partial f}{\partial w_1} = \left[-\frac{1}{2}(w_1^2 + w_2^2)^{-3/2} \cdot 2w_1 \right] e^{-(w_1 b)^2} + (w_1^2 + w_2^2)^{-1/2} \left[e^{-(w_1 b)^2} \cdot (-2w_1 b^2) \right]$
 $f_{w_1} = -w_1 e^{-(w_1 b)^2} \left[\frac{1}{(w_1^2 + w_2^2)^{3/2}} + \frac{2b^2}{(w_1^2 + w_2^2)^{1/2}} \right]$
2. **Partial w.r.t b :** $f_b = (w_1^2 + w_2^2)^{-1/2} \cdot e^{-(w_1 b)^2} \cdot (-2w_1^2 b)$ **Analysis:** The negative sign in f_b indicates that as bias b increases, the margin f decreases. The presence of $e^{-(w_1 b)^2}$ means that for large b , the gradient approaches zero, representing a "vanishing gradient" where the model stops learning.

Problem 5: Trajectory Tracking in Robotics A robotic arm's end-effector energy potential is $E(x, y) = \sqrt{x^2 + y^3}$. The arm follows a spiral trajectory defined by $x = e^t \cos(2t)$ and $y = e^t \sin(2t)$.

- Apply the multivariable chain rule to find the power (rate of change of energy) $\frac{dE}{dt}$ at $t = 0$.
- **Scenario Analysis:** Does the energy of the system dissipate or accumulate as the arm moves away from the origin at $t = 0$?

Function: $E = (x^2 + y^3)^{1/2}$ where $x = e^t \cos 2t$, $y = e^t \sin 2t$

Step-by-Step Solution:

1. Inner values at $t = 0$: $x = e^0 \cos 0 = 1$; $y = e^0 \sin 0 = 0$.

2. Outer Partial at $(1, 0)$:

- $E_x = \frac{1}{2}(x^2 + y^3)^{-1/2}(2x) = \frac{1}{\sqrt{1+0}}(1) = 1$
- $E_y = \frac{1}{2}(x^2 + y^3)^{-1/2}(3y^2) = 0$ (since $y = 0$)

3. Inner Derivatives at $t = 0$:

- $x' = e^t \cos 2t - 2e^t \sin 2t \rightarrow 1 - 0 = 1$
- $y' = e^t \sin 2t + 2e^t \cos 2t \rightarrow 0 + 2 = 2$

4. Chain Rule: $\frac{dE}{dt} = E_x \frac{dx}{dt} + E_y \frac{dy}{dt} = (1)(1) + (0)(2) = 1$.

Q:- In Binary Cross-Entropy loss, the error E depends on the model's prediction P . The prediction P is a result of a sigmoid activation fn applied to a weight w & input x .

$$E(P) = -\ln(P) \quad (\text{Simplified Loss})$$

$$P(w) = \frac{1}{1 + e^{-wx}} \quad (\text{Sigmoid Activation})$$

If input $x = 2$ & weight w is being updated as fn of time such that $w(t) = \ln(t^2 + 2)$, find rate of change of error wrt time at $t = 1$. What does answer represents in this scenario.

$$\text{Sol } \frac{dE}{dt} = \frac{dE}{dP} \cdot \frac{dP}{dw} \cdot \frac{dw}{dt}$$

$$\frac{dE}{dP} = -\frac{1}{P}, \quad \frac{dP}{dw} = \frac{x e^{-wx}}{(1 + e^{-wx})^2}, \quad \frac{dw}{dt} = \frac{2t}{t^2 + 2}$$

$$w(1) = \ln(1^2 + 2) = \ln(2) \approx 0.693$$

$$x = 2, \quad wx = 2 \ln(2) = \ln(4)$$

$$e^{-wx} = e^{-\ln(4)} = \frac{1}{4} = 0.25$$

$$P(1) = \frac{1}{1 + 0.25} = 0.8$$

Problem 6: Information Entropy Flux In a communication channel, the entropy H depends on two probabilities p and q such that $H(p, q) = -p \ln(p) - q \ln(q)$. These probabilities change over time as the signal-to-noise ratio (SNR) fluctuates: $p(s) = \frac{1}{1+s^2}$ and $q(s) = \frac{s^2}{1+s^2}$.

- Find the rate of change of entropy with respect to SNR, $\frac{dH}{ds}$.
- Scenario Analysis:** Find the value of s where the rate of change of entropy is zero. What does this represent in terms of the channel's stability?

Function: $H = -p \ln p - q \ln q$

Step-by-Step Solution:

- Find $\frac{dH}{ds}$: $\frac{dH}{ds} = \frac{\partial H}{\partial p} \frac{dp}{ds} + \frac{\partial H}{\partial q} \frac{dq}{ds}$
 - $\frac{\partial H}{\partial p} = -(1 \cdot \ln p + p \cdot \frac{1}{p}) = -(\ln p + 1)$
 - Similarly, $\frac{\partial H}{\partial q} = -(\ln q + 1)$
- Calculate dp/ds and dq/ds :
 - $p = (1 + s^2)^{-1} \implies p' = -(1 + s^2)^{-2}(2s)$
 - $q = \frac{s^2}{1+s^2} \implies q' = \frac{(1+s^2)(2s) - s^2(2s)}{(1+s^2)^2} = \frac{2s}{(1+s^2)^2}$
- Set $\frac{dH}{ds} = 0$: $-(\ln p + 1)[\frac{-2s}{(1+s^2)^2}] - (\ln q + 1)[\frac{2s}{(1+s^2)^2}] = 0$
 $\frac{2s}{(1+s^2)^2} [(\ln p + 1) - (\ln q + 1)] = 0 \implies \ln p = \ln q \implies p = q$
- Solve for s : $\frac{1}{1+s^2} = \frac{s^2}{1+s^2} \implies s^2 = 1 \implies s = 1$.

$$\frac{dE}{dp} = \frac{-1}{0.8} = -1.25$$

$$\frac{dp}{dw} = \frac{2(0.25)}{(1.25)^2} = 0.32$$

$$\frac{dw}{dt} = \frac{2(1)}{1^2 + 1} = 1$$

$$\frac{dE}{dt} = (-1.25) \cdot (0.32) \cdot (1) = -0.4$$

Negative value here indicates that as time progresses, the error is decreasing.

Problem 7: Composite Function Optimization Let $z = f(u, v) = \ln(u^2 + v^2)$, where $u = r \cos \theta$ and $v = r \sin \theta$.

- Use the chain rule to find $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$.
- **Scenario Analysis:** Show that z is independent of θ . Explain geometrically why the potential of a point depends only on its distance r from the origin in this specific system.

$$z = \ln(u^2 + v^2) \text{ where } u = r \cos \theta, v = r \sin \theta$$

Step-by-Step Solution:

$$1. \text{ Partial w.r.t } r: \frac{\partial z}{\partial r} = \frac{2u}{u^2 + v^2} (\cos \theta) + \frac{2v}{u^2 + v^2} (\sin \theta) \text{ Substitute } u^2 + v^2 = r^2: \frac{\partial z}{\partial r} = \frac{2r \cos \theta}{r^2} \cos \theta + \frac{2r \sin \theta}{r^2} \sin \theta = \frac{2r(\cos^2 \theta + \sin^2 \theta)}{r^2} = \frac{2}{r}$$

$$2. \text{ Partial w.r.t } \theta: \frac{\partial z}{\partial \theta} = \frac{2u}{r^2} (-r \sin \theta) + \frac{2v}{r^2} (r \cos \theta) \frac{\partial z}{\partial \theta} = \frac{-2r^2 \cos \theta \sin \theta + 2r^2 \sin \theta \cos \theta}{r^2} = 0$$

Conclusion: Since $z_\theta = 0$, the function's value depends solely on the radius r , making it rotationally invariant.